TECHNICAL DETAILS FOR 1.1 KV SINGLE CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Nominal Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
						With Al Conductor	With Cu Conductor
	Aluminium	Copper				AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	1.80	8	80	100
6	1/3	1/7	1.00	1.80	9	90	120
10	1/7	1/7/6	1.00	1.80	10	110	170
16	6	6	1.00	1.80	10	150	250
25	6	6	1.20	1.80	12	200	350
35	6	6	1.20	1.80	13	250	450
50	6	6	1.40	1.80	15	300	600
70	12	12	1.40	1.80	16	400	800
95	15	15	1.60	1.80	18	500	1050
120	15	18	1.60	2.00	20	600	1300
150	15	18	1.80	2.00	22	700	1600
185	30	30	2.00	2.00	24	850	1950
240	30	34	2.20	2.00	27	1100	2500
300	30	34	2.40	2.00	30	1300	3100
400	53	53	2.60	2.20	34	1700	3950
500	53	53	3.00	2.20	37	2100	5000
630	53	53	3.40	3.40	42	2700	6450
800	53	53	3.40	2.40	46	3250	8050
1000	53	53	3.40	2.60	50	3950	9950

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper asper Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4Sq.mm, 6Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted Circular
   For Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm and above - Stranded Compacted Circular
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Black, Red
3. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 °C		Approx. Conductor A.C. Resistance at 70 °C		Reactance of Cable at 50Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.140	0.580	36	33	32	46	42	43	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.127	0.680	44	42	41	57	54	54	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.118	0.830	59	56	56	75	72	72	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.110	1.010	75	71	72	94	92	92	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.105	1.050	97	93	99	125	120	125	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.100	1.220	120	110	120	150	140	155	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.098	1.220	145	130	150	180	165	190	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.091	1.430	170	155	185	220	200	235	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.088	1.470	205	180	215	265	230	275	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.086	1.620	230	200	240	300	255	310	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.085	1.620	265	220	270	340	280	345	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.084	1.620	300	240	305	380	305	390	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.082	1.720	335	270	350	420	340	445	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.080	1.740	370	295	395	465	370	500	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.080	1.810	410	335	455	500	405	510	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.079	1.860	435	355	490	540	430	610	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.077	1.870	485	395	560	590	465	680	47.900	72.500
800	0.0367	0.0221	0.0503	0.0303	0.077	1.980	525	420	650	645	500	745	60.800	92.000
1000	0.0291	0.0176	0.0422	0.0255	0.076	2.200	570	445	735	705	546	890	76.000	115.000

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV TWO CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	13	230	280
6	1/3	1/7	1.00	0.3	1.80	15	270	350
10	1/7	1/7/6	1.00	0.3	1.80	16	330	480
16	6	6	1.00	0.3	1.80	16	350	500
25	6	6	1.20	0.3	2.00	19	450	750
35	6	6	1.20	0.3	2.00	20	550	950
50	6	6	1.40	0.3	2.00	23	700	1250
70	12	12	1.40	0.3	2.00	25	850	1650
95	15	15	1.60	0.4	2.20	29	1150	2250
120	15	18	1.60	0.4	2.20	31	1300	2700
150	15	18	1.80	0.4	2.40	33	1600	3300
185	30	30	2.00	0.5	2.40	36	1900	4100
240	30	34	2.20	0.5	2.60	42	2450	5250
300	30	34	2.40	0.6	2.80	45	2950	6500
400	53	53	2.60	0.7	3.20	51	3800	8300
500	53	53	3.00	0.7	3.40	57	4750	10550
630	53	53	3.40	0.7	3.80	64	6000	13500

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm and above - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	32	27	27	41	35	35	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	40	34	35	50	44	45	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	55	45	47	70	58	60	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	70	58	59	90	75	78	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	90	76	48	115	97	105	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	110	92	99	140	125	125	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	135	115	125	165	145	155	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	160	140	150	205	180	195	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	190	170	185	240	215	230	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	210	190	210	275	235	265	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	240	210	240	310	270	305	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	275	240	275	350	300	350	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	320	275	325	405	345	410	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	355	305	365	450	385	465	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	385	345	420	490	425	530	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	425	380	475	540	460	605	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	465	415	540	640	550	785	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	14	250	330
6	1/3	1/7	1.00	0.3	1.80	15	310	420
10	1/7	7/6	1.00	0.3	1.80	17	370	580
16	6	6	1.00	0.3	1.80	18	450	700
25	6	6	1.20	0.3	2.00	21	600	1050
35	6	6	1.20	0.3	2.00	23	700	1350
50	6	6	1.40	0.3	2.00	26	900	1750
70	12	12	1.40	0.3	2.00	29	1200	2400
95	15	15	1.60	0.4	2.20	33	1500	3200
120	15	18	1.60	0.4	2.20	36	1800	3900
150	15	18	1.80	0.4	2.40	40	2200	4800
185	30	30	2.00	0.5	2.40	43	2700	5950
240	30	34	2.20	0.5	2.80	49	3550	7750
300	30	34	2.40	0.6	3.00	54	4200	9600
400	53	53	2.60	0.7	3.40	61	5350	12200
500	53	53	3.00	0.7	3.60	69	6750	15500
630	53	53	3.40	0.7	4.00	77	8550	19900

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular,
   16 Sq. mm and above - Stranded Compacted shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm
   - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 °C		Approx. Conductor A.C. Resistance at 70 °C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	28	23	23	36	30	30	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	35	30	30	45	38	39	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	46	39	40	60	50	52	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	60	50	51	77	64	66	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	110	95	105	145	125	135	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	22	265	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	335	290	375	425	360	455	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE & HALF CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
3x25+16	6/6	6/6	1.20/1.00	0.3	2.00	23	700	1250
3x35+16	6/6	6/6	1.20/1.00	0.3	2.00	25	800	1550
3x50+25	6/6	6/6	1.40/1.20	0.3	2.20	28	1050	2050
3x70+35	12/6	12/6	1.40/1.20	0.4	2.20	32	1400	2800
3x95+50	15/6	15/6	1.60/1.40	0.4	2.20	36	1800	3700
3x120+70	15/12	18/12	1.60/1.40	0.5	2.40	39	2200	4700
3x150+70	15/12	18/12	1.80/1.40	0.5	2.40	43	2550	5550
3x185+95	30/15	30/15	2.00/1.60	0.5	2.60	47	3150	6900
3x240+120	30/15	34/18	2.20/1.60	0.6	3.00	53	4050	8950
3x300+150	30/15	34/18	2.40/1.80	0.6	3.20	58	4900	11100
3x400+185	53/30	53/30	2.60/2.00	0.7	3.40	64	6150	14000
3x500+240	53/30	53/34	3.00/2.20	0.7	3.80	76	7900	18050
3x630+300	53/30	53/34	3.40/2.40	0.7	4.00	84	9900	22950

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted Shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow, Blue & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
3x25+16	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
3x35+16	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
3x50+25	0.6410	0.3870	0.7690	0.4640	0.082	0.49	11	95	105	145	125	135	3.800	5.750
3x70+35	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
3x95+50	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
3x120+70	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
3x150+70	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	225	265	11.400	17.300
3x185+95	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
3x240+120	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.600
3x300+150	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
3x400+185	0.0778	0.0470	0.0759	0.0580	0.072	0.70	335	290	375	425	350	455	30.400	46.000
3x500+240	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
3x630+300	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	15	290	400
6	1/3	1/7	1.00	0.3	1.80	17	350	510
10	1/7	1/6/7	1.00	0.3	1.50	19	440	710
16	6	6	1.00	0.3	2.00	21	550	950
25	6	6	1.20	0.3	2.00	23	750	1350
35	6	6	1.20	0.3	2.00	26	900	1700
50	6	6	1.40	0.4	2.20	29	1200	2300
70	12	12	1.40	0.4	2.20	32	1500	3100
95	15	15	1.60	0.4	2.40	37	2000	4200
120	15	18	1.60	0.5	2.40	41	2400	5150
150	15	18	1.80	0.5	2.60	45	2900	6350
185	30	30	2.00	0.6	2.80	50	3600	7900
240	30	34	2.20	0.6	3.00	56	4550	10200
300	30	34	2.40	0.7	3.40	64	5650	12750
400	53	53	2.60	0.7	3.60	70	7000	16100
500	53	53	3.00	0.7	4.00	79	8950	20550
630	53	53	3.40	0.7	4.00	89	11200	26250

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular,
   16 Sq. mm and above - Stranded Compacted shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular,
   10 Sq. mm - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow & Blue
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	28	23	23	36	30	30	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	35	30	30	45	38	39	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	46	39	40	60	50	52	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	60	50	51	77	64	66	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	110	95	105	145	125	135	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	22	265	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	335	290	375	425	360	455	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.